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State Finished

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Time taken 6 mins 25 secs

Grade 10.00 out of 10.00 (100%)

Question 1

Correct

Mark 2.50 out of 2.50

For the system $AX = b$ where:

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & -1 & 3 \\ 0 & 2 & 5 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 12 \\ 17 \end{bmatrix}$$

We want to solve $AX = b$ using LU factorization without pivoting: $A = LU$, then solve $Ly = b$ then $Ux = y$.

The correct L matrix is

- ☐ A. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & \frac{2}{3} & 4 \end{bmatrix}$
- ☐ B. $\begin{bmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix}$
- ☒ C. $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix}$ ✓
- ☐ D. $\begin{bmatrix} 2 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & \frac{2}{3} & 1 \end{bmatrix}$

Your answer is correct.

The correct answer is: $\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 0 & -\frac{2}{3} & 1 \end{bmatrix}$

Question 2

Correct

Mark 2.50 out of 2.50

For the system $AX = b$ where:

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & -1 & 3 \\ 0 & 2 & 5 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 12 \\ 17 \end{bmatrix}$$

We want to solve $AX = b$ using LU factorization without pivoting: $A = LU$, then solve $Ly = b$, then $Ux = y$.**d-Solving $Ux = y$, the final solution x is:**

- ☐ A. $\begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}$
- ☒ B. $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$ ✓
- ☐ C. $\begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$
- ☐ D. $\begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix}$

Your answer is correct.

The correct answer is: $\begin{bmatrix} 1 \\ 1 \\ 3 \end{bmatrix}$ **Question 3**

Correct

Mark 2.50 out of 2.50

For the system $AX = b$ where:

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & -1 & 3 \\ 0 & 2 & 5 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 12 \\ 17 \end{bmatrix}$$

We want to solve $AX = b$ using LU factorization without pivoting: $A = LU$, then solve $Ly = b$, then $Ux = y$.**C-Solving $Ly = b$, the intermediate vector y equals:**

- ☐ A. $\begin{bmatrix} 3 \\ 0 \\ 13 \end{bmatrix}$
- ☐ B. $y = \begin{bmatrix} 3 \\ -6 \\ 21 \end{bmatrix}$
- ☐ C. $\begin{bmatrix} 3 \\ 12 \\ 17 \end{bmatrix}$
- ☒ D. $y = \begin{bmatrix} 3 \\ 6 \\ 21 \end{bmatrix}$ ✓

Your answer is correct.

The correct answer is: $y = \begin{bmatrix} 3 \\ 6 \\ 21 \end{bmatrix}$

Question 4

Correct

Mark 2.50 out of 2.50

For the system $AX = b$ where:

$$A = \begin{bmatrix} 2 & 1 & 0 \\ 4 & -1 & 3 \\ 0 & 2 & 5 \end{bmatrix}, b = \begin{bmatrix} 3 \\ 12 \\ 17 \end{bmatrix}$$

We want to solve $AX = b$ using LU factorization without pivoting: $A = LU$, then solve $Ly = b$, then $Ux = y$

a-The correct U matrix is:

- ☐ A. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 0 \\ 0 & 0 & 7 \end{bmatrix}$
- ☒ B. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 3 \\ 0 & 0 & 7 \end{bmatrix}$ ✓
- ☐ C. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 3 \\ 0 & 0 & 3 \end{bmatrix}$
- ☐ D. $\begin{bmatrix} 2 & 1 & 0 \\ 0 & 3 & 3 \\ 0 & 0 & 7 \end{bmatrix}$

Your answer is correct.

The correct answer is: $\begin{bmatrix} 2 & 1 & 0 \\ 0 & -3 & 3 \\ 0 & 0 & 7 \end{bmatrix}$